NTNU

Modern Machine Learning in Practice

Predicting Unit Commitment in the Tokke-Vinje Hydropower System

Group 26 (Kaggle group name: 26)

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1 Introduction

SINTEF is an independent research organization headquartered in Trondheim that researches hydropower, among other things (Store norske leksikon, n.d.). The Tokke-Vinje watercourse is a system comprising 14 generating units in 9 power plants connected to 17 reservoirs by a web of rivers in Norway's Tokke and Vinje municipalities. SINTEF published a dataset containing the unit commitment (UC) decisions of these generators from 2015 to 2024, together with the inputs used to make them: day-ahead prices, inflows, reservoir volumes, water values, environmental constraints and the topology of the system (SINTEF Energy Research, n.d.). The aim of this project is to predict the hourly on/off decisions of all 14 units over a 168-hour horizon (2352 binary targets per case) from these inputs, as a fast alternative to solving the optimization problem. Models are trained on the cases from 2015-01-01 to 2022-12-31 and evaluated on Kaggle on the cases from 2023-01-01 to 2024-12-25, using the micro-averaged ROC-AUC.

The data was first examined through exploratory data analysis (EDA) to understand the problem and guide feature engineering. Based on the domain knowledge gathered there, we built features that make the producer's decision rule explicit, including three small optimization models (a dispatch surrogate, a start/stop cost evaluation and a linear program of the whole cascade) whose outputs are used as features. Several types of predictors were tested (Random Forest (Breiman, 2001), XGBoost (Chen and Guestrin, 2016), AutoGluon (Erickson et al., 2020) with LightGBM (Ke et al., 2017), CatBoost (Prokhorenkova et al., 2018) and ExtraTrees (Geurts et al., 2006), and a temporal convolutional network (Bai et al., 2018)), and the two submitted models blend a tree-based ensemble with a neural network. Finally, the models were interpreted with SHAP (Lundberg and Lee, 2017; Lundberg et al., 2020), permutation importance (Breiman, 2001), partial dependence (Friedman, 2001) and surrogate trees (Molnar, 2022).

2 Exploratory Data Analysis

EDA was performed on both the data in the core dataset and the data in the extended dataset to gain an understanding and an overview of the problem as well as the relevant variables. Each variable was examined separately first and plotted as a function of time to visualize the data and possibly reveal anomalies, patterns, or trends to research further. This report contains a small selection of relevant plots to illustrate certain points discussed in the text. According to the dataset documentation, the unit commitment decisions were produced by SINTEF's SHOP optimization tool (SINTEF SHOP, n.d.), solved for a new 168-hour horizon every day with the price, inflow, volume, water value and constraint data as inputs; the targets are therefore the output of an optimizer, not of a human operator. Through the EDA, the data was determined not to be majorly corrupted or missing significant amounts of entries, and as a result, no further cleanup was performed. The only data anomaly found was that starting on 30.09.2025 from 22:00 to 22:15, the day-ahead price data changed from 1-hour to 15-minute intervals (NVE/RME, n.d.) (Figure 1, bottom). However, as the relevant timeframe of this project ranges from 01.01.2015 to 31.12.2024, our model is not affected by that, and we do not need to account for that. Additionally, according to the source of the data, "these datasets are further processed and quality-assured by SINTEF Energy Research" (SINTEF Energy Research, n.d.).

Prices were found to be quite volatile, and large spikes in later years were also discovered (Figure 1, top). These findings are in accordance with the factual spikes in energy prices recorded at that time as a result of geopolitical events and lower-than-average precipitation in southern Norway (Statistics Norway, n.d.; ScienceDirect, n.d.). Trends and seasonalities were also checked, resulting in the same peaks and suggesting the use of normalized and ranked features for training the model.

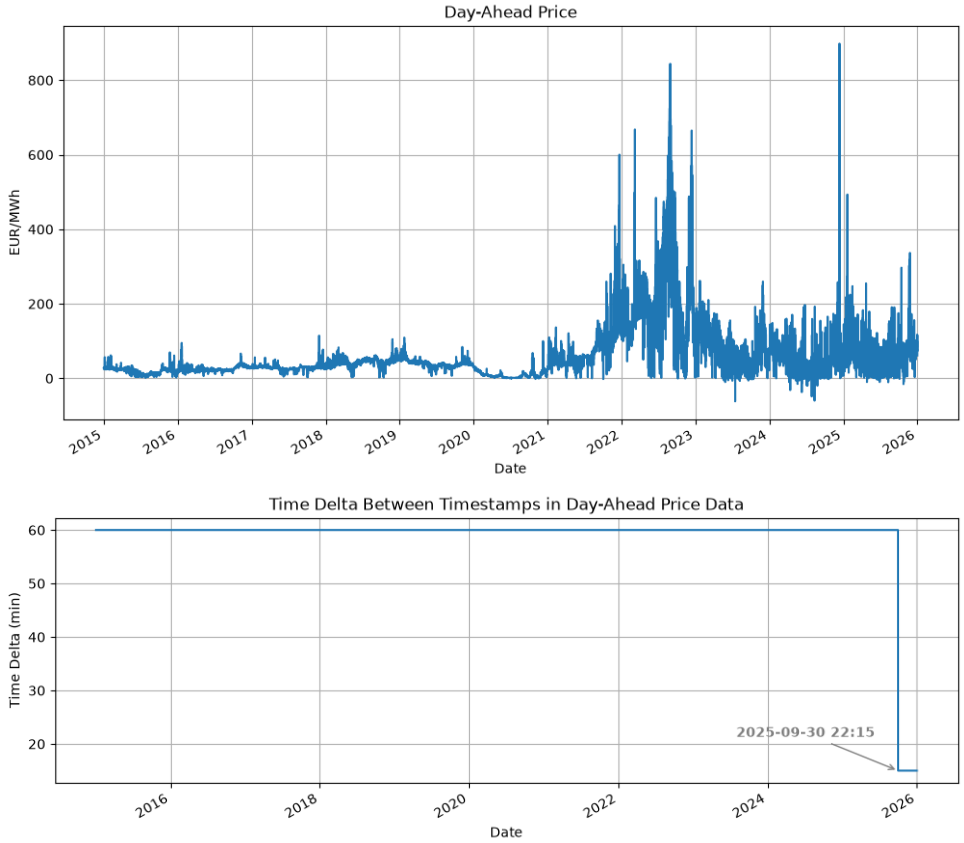


Figure 1: Top: hourly day-ahead price 2015–2025. Bottom: time step between consecutive price timestamps, showing the switch from 60 to 15 minutes on 30.09.2025.

Evidence of this lower-than-average precipitation was also found in the inflow data (2022 had a total inflow of approx. $29 \cdot 10^8\,\mathrm{m}^3$, while the average over 10 years was approx. $36 \cdot 10^8\,\mathrm{m}^3$), which is assumed to mainly reflect rainfall. The inflow data was examined per power plant, and a visual inspection of the plotted data seems to suggest yearly patterns shared by most reservoirs, although less pronounced in some, even though the average difference in absolute inflow between the minimum (Vatjern) and maximum (Songavatn) reservoirs is about 2 orders of magnitude. These patterns are likely a result of seasonal differences in precipitation (meteoblue, n.d.). Figure 2 only plots data from 2015 onwards to more sharply show the patterns. It will still be useful to incorporate earlier data into the model through lagged or ranked features. The plots also include the minimum flow requirements where they exist (not all reservoirs have one). The inflow often greatly exceeds these requirements, meaning that provisioning extra water storage will likely not be necessary to meet them, but still dips below them occasionally, suggesting the requirements might still be valuable to take into account. Lastly, a correlation matrix between the reservoirs with respect to their inflows was computed, and the inflows of some reservoirs were found to be perfectly correlated, implying they are directly up/downstream from each other and share the same micro-climate basin, which is confirmed by the topology.

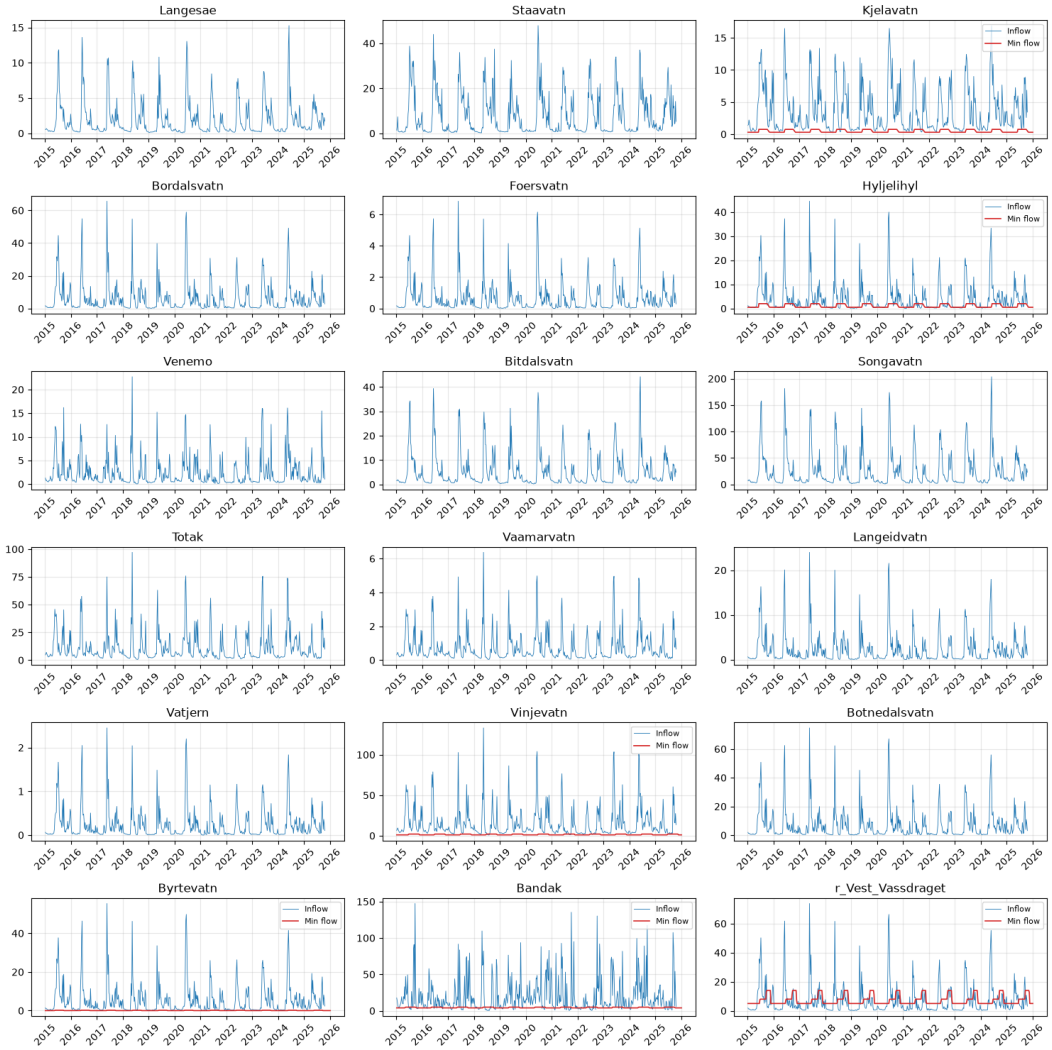


Figure 2: Weekly mean inflow per reservoir (2015 onwards), with the minimum flow requirement in red where one exists.

The volume data was also considered per power plant and visualized as such (Figure 3). These plots also reveal clear patterns, although there are many more variations and deviations than with the inflow data. Most reservoirs seem to again indicate some seasonality, with the exception of Vinjevatn and Bandak, which only show weak seasonality, which is true to reality as well (NVE Sildre, n.d.). The storage volume of two reservoirs, Hyljelihyl and Vatjern, also remains unchanged throughout the entire dataset, which seems to be correct (NVE Vannkart, n.d.) and implies that no water is stored at these reservoirs; everything flowing in also directly flows back out. Three reservoirs also have minimum volume requirements, which were added to their respective plots. These requirements are also clearly seasonal and align more closely with the actual storage volumes, suggesting a significant effect on the storage volume and, as a result, the water management of those reservoirs as a whole. These minimum requirements are the result of a measure to mitigate the impact of the hydropower plants on the local ecology (Regjeringen, n.d.). A correlation matrix was again computed, and no perfect correlations were present this time, although some strong correlations were present. This can be partly explained by geographical proximity and up-/downstream connectedness, but might also imply that these reservoirs are managed using a common shadow price (and as a result turned on/off at mostly the same times).

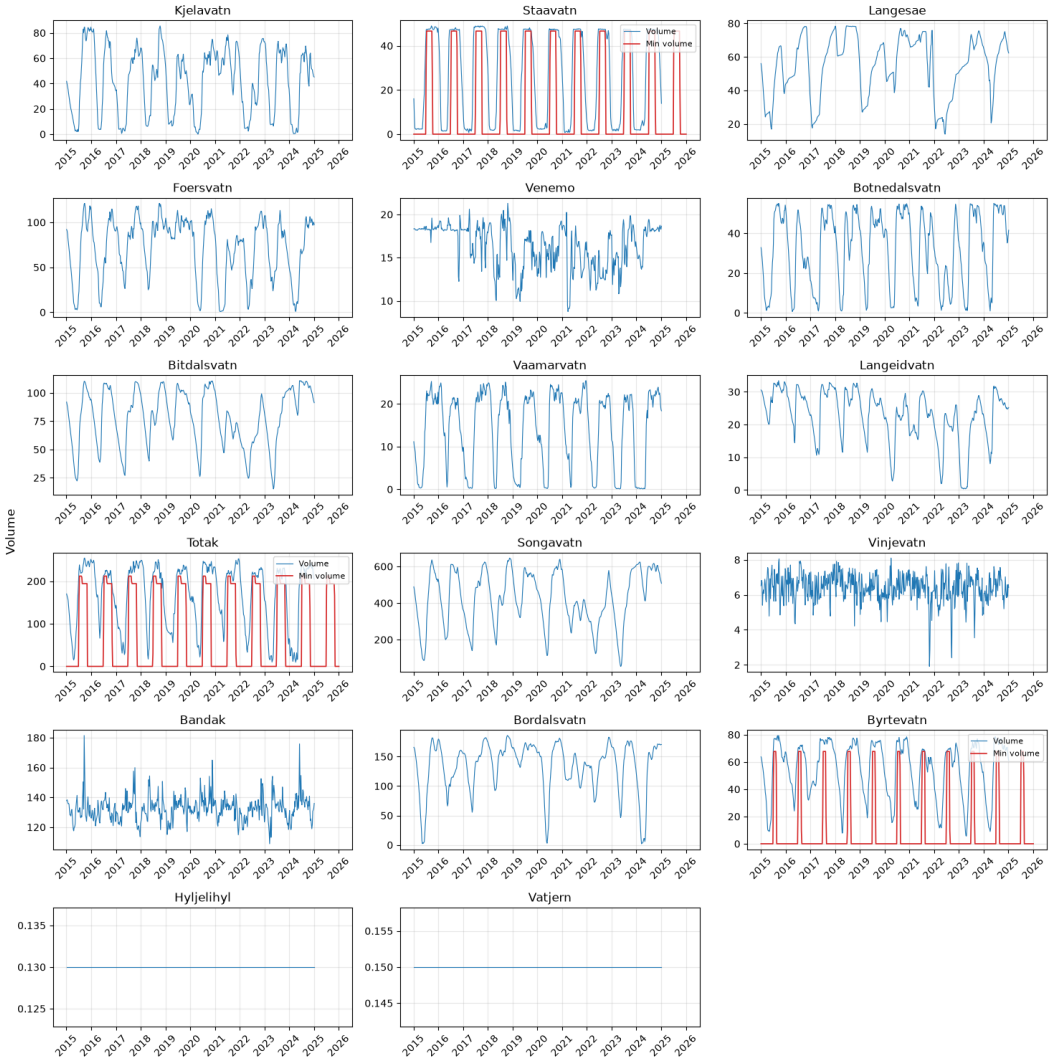


Figure 3: Weekly mean reservoir volume, with the minimum volume requirement in red for the three reservoirs that have one.

The synthetic water value is important to balance supply and demand of water. If producers set the water value too low, consumption increases, which can lead to empty reservoirs before the dry season is over. This value depends on how long it is possible to store the water and is calculated using statistics, forecasts, and models to make advanced mathematical calculations on production and consumption (NVE water values, n.d.). When plotted (Figure 4), it shows a very similar picture to the day-ahead price: it seems just as volatile and shows almost identical spikes in 2022. The average correlation between the price and water value per unit is 0.88, which makes a lot of sense intuitively, as the water value is directly linked to the electricity prices. The water value drops down to just above zero at certain points, which probably indicates an excess of water in those reservoirs. Each reservoir spends about 10% of the time in such a state of excess. The hypothesis of common shadow prices put forward in the previous paragraph also seems to be supported by the fact that several groups of plots have almost identical shapes.

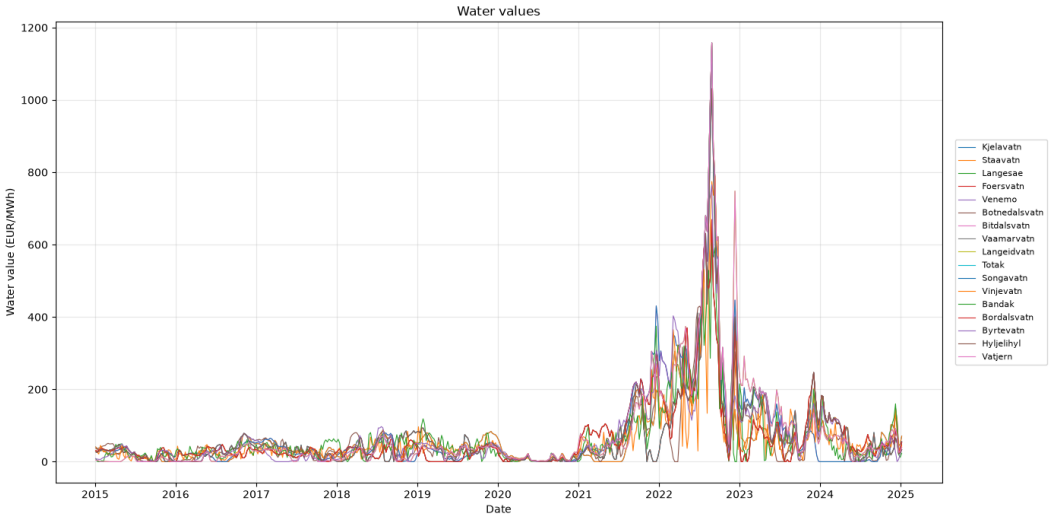


Figure 4: Weekly mean synthetic water value per reservoir.

The unit commitment decisions are stored in cases of 168-hour horizons for each of the 14 generating units. For a profit-maximizing producer, generators are expected to be turned on when prices are high. Contrary to this expectation, the median day-ahead price is lower in hours where a generator is committed than in hours where it is off (Figure 5, Table 1). This holds for 12 of the 14 units, with differences ranging from about -1 to -8 EUR/MWh (largest for Haukeli_G1, Lio_G1, Tokke_G1 and Tokke_G3). The median price when on is remarkably similar across units (about 30 EUR/MWh), while the median price when off varies more. This suggests that commitment is not driven by the spot price alone. Possible explanations are that the decisions come from an optimization with a 168-hour horizon, where the water value, reservoir levels, inflow and minimum flow constraints can outweigh hourly prices. Since water values are partly set by expected future prices, plants may run in moderately priced hours when reservoirs are full or inflow is high, and stay off in high-priced hours if the water is judged more valuable later. Seasonal effects may also play a role, since prices and operation both vary through the year. Overall, price alone is a weak predictor of unit commitment, which motivates informed feature extraction.

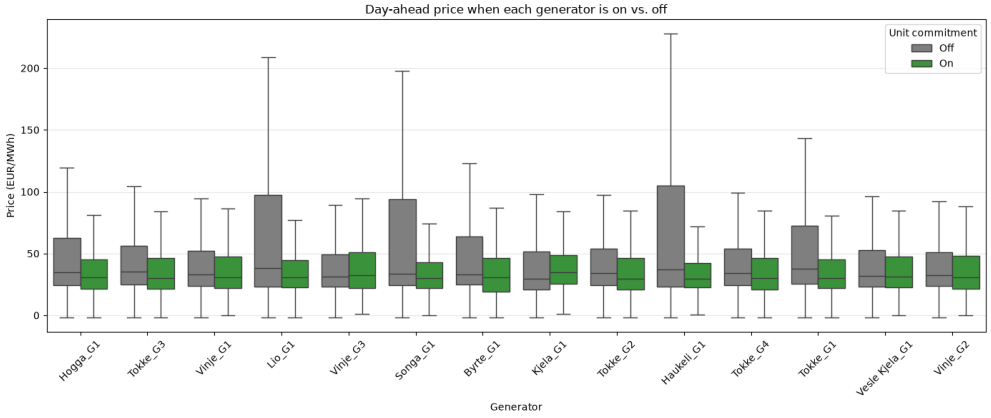


Figure 5: Day-ahead price in the hours in which each generator is off (grey) and on (green).

Table 1: Median day-ahead price (EUR/MWh) when each unit is off and on.

Unit	Off	On	Difference
Byrte_G1	33.0	30.8	−2.2
Haukeli_G1	37.3	29.4	−7.9
Hogga_G1	34.8	30.5	−4.3
Kjela_G1	29.4	34.8	5.5
Lio_G1	38.5	30.9	−7.6
Songa_G1	33.6	30.1	−3.5
Tokke_G1	37.9	30.1	−7.8
Tokke_G2	34.4	29.8	−4.6
Tokke_G3	35.4	30.0	−5.5
Tokke_G4	34.3	30.0	−4.2
Vesle Kjela_G1	32.2	31.1	−1.0
Vinje_G1	33.0	30.8	−2.3
Vinje_G2	32.8	30.7	−2.1
Vinje_G3	31.5	32.5	1.0

The topology file describes the physical structure of the watercourse as a directed graph of 17 reservoirs, 9 power plants with 14 generating units (1003 MW installed in total), 5 tunnels and 37 river segments (bypass, spill and regulated flow). Figure 6 shows the resulting network, with elevation on the vertical axis and the flow direction running from upstream (left) to downstream (right). It shows that the system is a strongly coupled cascade rather than a set of independent plants. The reservoirs span roughly 1000 m of elevation, from Langesae (HRL 1090 m) down to Bandak (73 m), and almost all water ends up in Bandak, either through the Tokke plant or through the river Vest-Vassdraget. The longest chain starts at Songavatn and passes through Songa, Totak, Vaamarvatn, Vinje, Vinjevatn, Tokke and Bandak before reaching Hogga. This means the discharge of an upstream plant becomes the inflow of the reservoir feeding the next plant, so that the commitment of one unit changes the water available to the units below it. Vinjevatn acts as a hub, receiving the discharge of Vinje and Haukeli and several river flows, and it feeds Tokke, the largest plant (4×107.5 MW). Reservoir sizes also differ strongly: Songavatn (640) is by far the largest, followed by Totak, Bordalsvatn and Bandak, while Hyljelihyl (0.26) and Vatjern (0.3) have almost no storage capacity. This agrees with the constant volumes found for these two reservoirs above, and it explains why they behave like pass-through points. The topology also supports the earlier observations that some reservoirs have perfectly correlated inflows and that several water values look alike, since reservoirs directly above or below each other share inflow patterns and probably the same marginal value of water. For feature engineering, this suggests that features of the upstream and downstream neighbours of a unit (such as their volumes, inflows or water values) may be informative in addition to features of the unit's own reservoir.

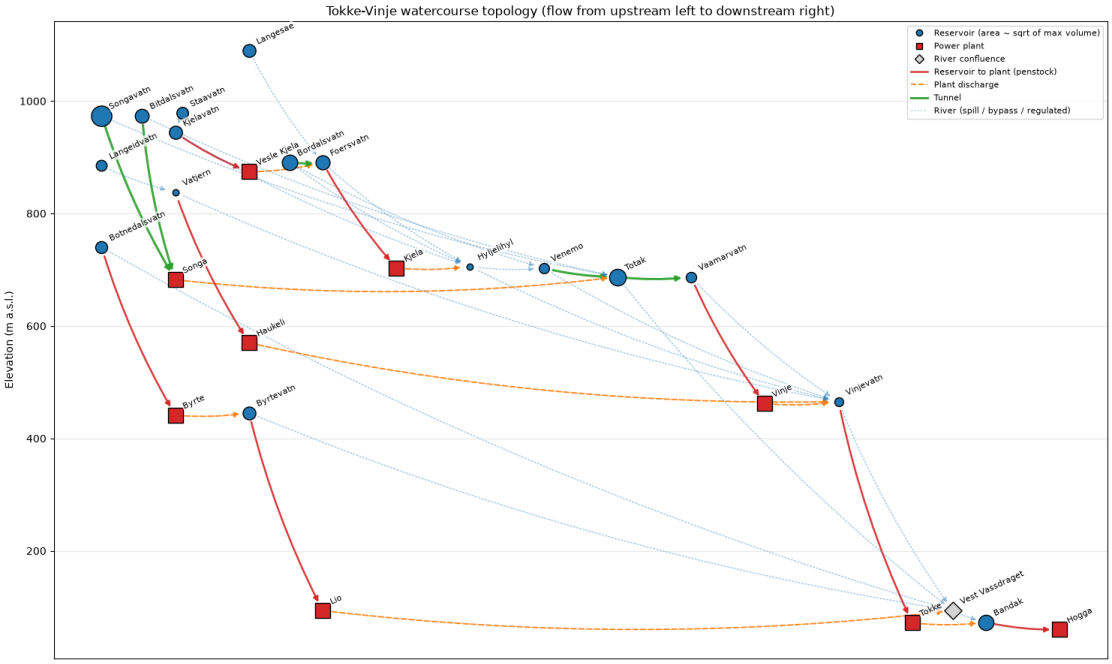


Figure 6: Topology of the Tokke-Vinje watercourse (17 reservoirs, 9 plants, 14 generators, 5 tunnels, 37 river segments; 1003 MW installed), with elevation on the vertical axis and flow from upstream (left) to downstream (right).

3 Feature Engineering (final feature set)

This section describes the features used by the two final models and how they were derived. Table 2 lists the feature groups of the final set and the version in which each was introduced; the full version history, including the options that were discarded, is described in Section 4.

Table 2: Feature groups of the final set (variant L) and the version that introduced them (see Table 4).

Feature group	Section	Introduced in
Calendar; price rank within week and day; inflow; volumes, filling, head and net head	3.3	v4 (C)
Upstream predicted commitment (stacking)	3.5	v4-B
Optimizer surrogate and expected upstream release	3.4	v9 (S)
Environmental constraints and constraint-corrected surrogate budget	3.3, 3.4	v11 (K)
Scale-free price and water value features; cumulative inflow and spill pressure; Haukeli reservoir mapping	3.3	v12 (N)
Unit physics, start/stop costs (Viterbi margins), feeder reservoirs	3.3, 3.4	v14 (P)
Cascade LP (relative flow, reduced cost)	3.4	v17 (L)

The EDA showed that the raw variables relate to UC decisions in complex and often indirect ways, so the information needed to explain the commitment decisions has to be made explicit through feature engineering. Both techniques were used: **feature extraction**, to create new variables by combining, transforming or aggregating columns (and, in the final models, by solving small optimization problems), and **feature selection**, to reduce the candidate set to variables that carry useful information, which limits redundancy, lowers the risk of overfitting and keeps the models interpretable.

3.1 Candidate features from domain logic

Rather than generating features blindly, knowledge of the operation of hydropower producers was used to come up with candidates that reflect the underlying decision problem. A reservoir-based producer does not simply run when the price is high. It compares the current price with the value of storing the water for later, and it is restricted by physical and regulatory limits. This led to the following groups of candidates:

- **Economic signals.** The difference or ratio between the day-ahead price and the water value is the most direct expression of the producer's decision rule: produce when the price exceeds the value of stored water. We also considered the price relative to its recent average, and its rank within the day or week, to capture whether an hour is cheap or expensive in context.
- **Temporal patterns.** Hour of day, day of week, and month capture the daily load pattern and the seasonal cycle. Lagged and rolling-window versions of the price (e.g., previous 24 hours, 7-day mean) capture short-term market dynamics.
- **Reservoir state and constraints.** The headroom above the minimum volume (volume − min_volume), the volume as a fraction of its historical range, and the change in volume over recent days describe how much flexibility a unit has. Likewise, inflow minus minimum flow indicates whether the flow constraint is binding. Cumulative or lagged inflow reflects the delayed response of reservoirs to snowmelt and rain.
- **System and plant interactions.** Generators in the same cascade or power station tend to be operated together, and due to the connectedness of the system, reservoirs and plants that are upstream/downstream from each other can significantly influence outcome decisions.

Throughout the development of the candidates, relative features (differences, ratios, ranks, and normalised values) were favoured over raw absolute values for three reasons. They are robust to changes over time: price levels shifted considerably over 2015–2024, so a threshold learned from absolute prices in the training years may not hold in the test years. They are comparable across units, which matters because reservoirs differ greatly in size and plants in capacity. And they match the decision logic: a producer does not compare the price to a fixed number, but to the value of the stored water.

3.2 Feature selection at the level of feature groups

No univariate filter was applied to individual features. Scoring features one at a time would miss interactions, and the tree-based learners are robust to redundant or uninformative columns, which they simply do not split on. Feature selection was instead done at the level of feature groups, by adding, removing or reformulating whole groups and comparing the resulting models.

As a rule, a group was kept only if it improved the local test and out-of-fold scores (Section 4.4). The result is a single feature set, called **variant L**, which feeds both final models. All features are computed per plant from inputs known at the start of the week. None of them uses the labels of the case being predicted; the only features derived from labels are the stacking features (Section 3.5), which are out-of-fold predictions learned from other training cases.

We also checked how informative three scores are when used alone against the unit commitment (Table 3). The price rank within the week, which only says which hours are the cheapest or most expensive, has a micro AUC of 0.589. The log of the price divided by the water value reaches 0.872. Comparing the price with the value of the stored water therefore carries far more information about the commitment than the price pattern alone.

Table 3: Micro AUC of single scores used alone against the unit commitment (all cases, 2015–2022).

Score used alone	Micro AUC
Price rank within the week	0.589
log(price/water value)	0.872
Optimizer surrogate (estimated running units)	0.777

3.3 Market, state and calendar features

- **Calendar:** hour of the horizon (0–167), hour of day (0–23), day of week (0–6) and day of year. They are kept although their relationship with the target is not monotonic: a unit may be on during the day and in the evening but off at night.
- **Price shape:** the log of the price relative to the weekly mean (also shifted one hour forward and backward), the rank of the price within the week and within the day, and z-scores of the price relative to the daily and weekly means.
- **Price versus water value:** the log of the price divided by the water value at the start and at the end of the week, for each upstream and downstream reservoir of the plant. This is the direct implementation of the producer's decision rule.
- **Reservoir and hydraulic state:** initial volumes (absolute and as a fraction of the maximum volume), log water values at the start and end of the week, upstream water level (head), net head after the downstream level and the turbine outlet line, the log water value difference between upstream and downstream reservoirs, and the normalised volume available in secondary feeder reservoirs. Absolute values such as volume and inflow were kept next to the relative features, because relative features discard level information. For Haukeli, Langeidvatn is used as the main upstream reservoir, although the plant draws its water from Vatjern: the volume of Vatjern is constant in the data, so the head and net head computed from it were constant and carried no information (the *Haukeli topology fix* of v12). The discharge capacity of Haukeli is still computed from the head at Vatjern.
- **Inflow and spill pressure:** hourly and weekly mean inflow into the upstream reservoirs, the cumulative inflow divided by the plant capacity, and the volume plus cumulative inflow minus the maximum volume, divided by the capacity. The latter measures how close a reservoir is to spilling, which forces the plant to run regardless of the price.
- **Environmental constraints:** the active minimum flow and minimum volume constraints relevant to the plant, the upstream volume above the minimum volume, the downstream volume deficit relative to its minimum volume, and, for Lio, the flow deficit in the river Vest-Vassdraget that the plant has to cover. Added as plain columns (v5, v6), these features did not improve the score; they helped only from v11 onwards, when the minimum volume was also used to correct the water budget of the surrogate (Section 3.4).

Up to v11, the price features were absolute (the price itself, the price minus the daily and weekly means, the price minus the water value) and the water values entered without a log. In v12 they were replaced by the scale-free versions listed above. The local test period reaches prices of 577 EUR/MWh, while the 99th percentile in the training years is 62 EUR/MWh, and tree splits learned on absolute levels cannot extrapolate to such values. This change, which follows the preference for relative features stated in Section 3.1, was one of the largest gains of the development (variant N in Table 5).

3.4 Optimizer-based features

The most distinctive step was to turn small optimization problems into features, so that the learners receive an approximation of the scheduling decision instead of having to discover it from raw columns. Price features tell a model which hours are the best, but not how many hours a plant will run in a week, which depends on the available water, the turbine capacity and the water value. An error analysis of the model without these features (v4-C) showed that most of the remaining error was at this weekly level ("how many hours", not "which hours").

Optimizer surrogate (per plant)

The surrogate approximates the dispatch of one plant over the 168-hour horizon without solving the full unit commitment problem. First, the maximum turbine discharge c_p (Mm^3/h) is estimated from the nominal power $P_{p,\mathrm{nom}}$ (MW) given in the topology, the hydraulic head H_p (m), and an assumed efficiency of $\eta = 0.88$:

$$H_p = \max\left(h_{\mathrm{up}} - \max(h_{\mathrm{down}}, h_{\mathrm{outlet}}),\, 5\right), \qquad c_p = \frac{P_{p,\mathrm{nom}}}{2.725 \cdot \eta \cdot H_p}. \tag{1}$$

Here, h_{up} and h_{down} denote the upstream and downstream water levels, and h_{outlet} the turbine outlet level (all in metres). The factor $2.725 = 9.81 \cdot 277.8/1000$ converts $\rho g Q H$ to MW when Q is expressed in Mm^3/h.

The available water volume $V_{\mathrm{available}}$ (Mm^3) is calculated from the initial storage V_0 (Mm^3), the natural inflows $Q_{\mathrm{in},t}$ (m^3/s), and the volume released by the upstream plants V_{upstream} (Mm^3):

$$V_{\mathrm{available}} = V_0 + 0.0036 \sum_{t=1}^{168} Q_{\mathrm{in},t} + V_{\mathrm{upstream}}. \tag{2}$$

The factor 0.0036 converts one hour of flow in m^3/s to Mm^3. The weekly water budget W (Mm^3) is bounded by

$$W_{\max} = \mathrm{clip}\left(V_{\mathrm{available}} - V_{\mathrm{required}},\, 0,\, 168 c_p\right), \tag{3}$$

$$W_{\min} = \min\left(\max(V_{\mathrm{available}} - V_{\mathrm{storage,max}},\, 0),\, W_{\max}\right), \tag{4}$$

where V_{required} is the minimum volume requirement and $V_{\mathrm{storage,max}}$ the maximum reservoir capacity. $W_{\min}$ is the release needed to avoid spilling, $W_{\max}$ the maximum usable volume, and $168 c_p$ the volume used if the plant ran at full capacity for the whole week.

The economically desirable consumption is estimated from the hours in which the day-ahead price P_t exceeds the water value λ (both in EUR/MWh):

$$W_{\mathrm{econ}} = \mathrm{clip}\left(c_p \sum_{t=1}^{168} \mathbf{1}(P_t > \lambda),\, W_{\min},\, W_{\max}\right). \tag{5}$$

Finally, the budget is allocated to the highest-priced hours:

$$\hat{u}_{p,t}(W) = n_p \cdot \mathrm{clip}\left(\frac{W}{c_p} - r_t,\, 0,\, 1\right), \tag{6}$$

where $\hat{u}_{p,t}$ is the plant-level dispatch indicator, n_p the number of units in plant p, and r_t the zero-based descending price rank of hour t (0 = highest price, 167 = lowest). Four dispatch profiles are generated, for $W_{\min}$, $W_{\max}$, and W_{econ} computed with the water value at the start and at the end of the week. The profile based on the final water value, $S_{2,p,t}$, is adjusted for each unit as $S_{p,j,t} = S_{2,p,t} - j$, where j is the zero-based unit index within the plant.

The surrogate's weekly budget follows the real share of running hours per week, which the price rank cannot do because its weekly mean is the same in every week. Across non-overlapping weeks, the correlation between the budget and the real share of ON hours is high for Byrte ($r = 0.99$), Vesle Kjela (0.97) and Songa (0.95), still good for Kjela (0.91) and Hogga (0.88), and weaker for Lio (0.70), Haukeli (0.62), Vinje (0.50) and Tokke (0.44), where the water coming from upstream is harder to approximate. Used alone, the surrogate reaches a micro AUC of 0.777 over all labelled cases (Table 3). From v11 onwards (variant K), V_{required} is the active minimum volume constraint, so the budget bounds account for the environmental constraints; before, it was zero.

The surrogate is used for every plant in the final set. When it was introduced (v9), it lowered the test AUC of Songa (0.964 to 0.938), so in v10 and v11 it was left out for that plant. After the switch to scale-free price features in v12, it improved Songa as well (0.971 with variant N), and the exception was dropped.

Start/stop cost evaluation (Viterbi margins, per unit)

These features, together with the relative marginal capacity α_u defined below, form the group called *unit physics* in Section 6. Earlier models switched units on and off much more often than the real schedules, in which units often stay on for several days, because starts and stops have a cost. To encode this persistence without using the target, a two-state forward-backward (Viterbi (Viterbi, 1967)) pass computes, for each unit and hour, the best weekly value with the unit ON minus the best weekly value with the unit OFF. The hourly margin of running unit u is

$$m_{u,t} = \alpha_u P_t - \theta, \tag{7}$$

where P_t is the day-ahead price, α_u the relative marginal capacity of the unit (its marginal power, after the main and penstock head losses caused by the units already running in the same plant, divided by its reference power), and θ a water value threshold. Each change of state is penalised by

$$h_u = \frac{C_{\text{start},u} + C_{\text{stop},u}}{2P_{\text{ref},u}}, \tag{8}$$

with the start and stop costs taken from the topology and $P_{\text{ref},u}$ the reference output of the unit (MW) at peak discharge, maximum reference head and peak efficiency. Three variants of the margin are computed, with w the final water value of the source reservoir and λ the water value shifted only when the weekly budget $[W_{\min}, W_{\max}]$ requires it:

1. **Base margin:** $P_t - w$ ($\alpha_u = 1$).
2. **Marginal capacity margin:** $\alpha_u P_t - w$.
3. **Budget-constrained margin** (`vit_budget`): $\alpha_u P_t - \lambda$.

The normalised margins, their sign (ON/OFF indicator), α_u and the shift $\log \lambda - \log w$ are used as features.

Cascade linear program (LP)

According to the data documentation, the decisions were generated with SINTEF's SHOP tool, a mixed-integer optimization of the whole system. To comply with the competition rules (no external tools or data), we did not use SHOP, but formulated a simplified continuous LP of the same system and solved it with the open-source HiGHS solver (Huangfu and Hall, 2018; HiGHS, n.d.) available in SciPy (Virtanen et al., 2020). While the surrogate looks at one plant at a time and only approximates the water coming from upstream, the LP computes the optimal dispatch of the entire cascade over the 168-hour horizon.

The LP has variables for every generating segment, river and tunnel flow and reservoir volume, plus slack variables for the environmental constraints, which are penalised heavily. The objective balances the income from generation against the value of the water used (its water value), and mass balances and capacity limits are enforced at every node. It has no on/off variables and no start/stop costs, so its outputs are soft signals rather than exact answers. Two outputs are used per unit:

- **Relative flow:** the optimal discharge divided by the maximum discharge of the unit segment, between 0 (the LP recommends not running) and 1 (full capacity).
- **Reduced cost** (`lp_reduced_cost`): the dual value of the capacity limit of the segment, converted to EUR/MWh and divided by the water value. A high value means the system would gain from more capacity on that unit in that hour, i.e. a strong incentive to run; zero means the unit is not constrained.

On the local test weeks, the LP reduced cost alone reaches a micro AUC of 0.900, against 0.774 for the surrogate on the same weeks. The improvement is largest where the surrogate was weakest: for Tokke, the correlation of the weekly LP flow with the real share of running hours is 0.86, against 0.44 for the surrogate budget.

3.5 Cascade (stacking) features

Since plants share water, a plant's decisions depend on those upstream of it. For the tree-based models, the predicted commitment of the upstream plants (mean probability, lagged by 1 and 3 hours, and weekly mean; since v4-B) and the expected upstream release (since v9, where it enters the surrogate's water budget as V_{upstream}) (predicted probability $\times$ discharge capacity) become features of the downstream plant. These predictions are generated out-of-fold (Section 5.1), so no information leaks from the targets into the training features. The TCN does not need these features because it sees all plants at once.

4 Model Development: from Baseline to Final Models

4.1 Evaluation protocol: the local test set

The labels of the Kaggle test cases (2023–2024) are hidden, and the public leaderboard only scores part of them with a limited number of submissions, so it cannot be used to compare dozens of versions and settings. All development decisions were therefore made on a **local test set**: a held-out part of the labelled data that plays the role of unseen future data. To build it, the labelled cases from 1 January 2015 to 31 December 2022 (2922 daily cases of 168 hours) were split chronologically into a training set (first 80%) and a test set (last 20%). Because each case covers a whole week and consecutive cases overlap by six days, 7 cases are left out between the two sets: the last training case starts on 25 May 2021 (and covers the week until 1 June), and the test set starts with the case of 2 June 2021, so no hour belongs to both sets. All local scores in this report are micro ROC-AUC on this test set unless stated otherwise; out-of-fold (OOF) scores are computed on the training set with contiguous folds.

The models submitted to Kaggle were retrained on all labelled cases until 31 December 2022. In Version 18, the cases from 2021 onwards are weighted 2.0, because their market conditions are closer to those of the test years (in v12 this weighting raised the public score from 0.98769 to 0.98811). This choice was evaluated only on the public leaderboard, because the local training set contains only five months of 2021. Version 20 deliberately keeps all weights at 1.0. This diversifies the two submissions, so that a design choice that may not hold on the private leaderboard (here, the assumption that recent years are more representative) affects only one of them. The choice was

checked on the public leaderboard: Version 20 scored 0.99104 with the 2.0 weighting and 0.99101 without it, a negligible difference, so the unweighted version was kept as the second submission.

4.2 Data formatting for tree-based models vs. neural networks

- **Tree-based models:** a long tabular format per power plant, where every row is one hour of one generator. If a plant has several generators, their rows are stacked into one dataset with a column identifying the unit.
- **Neural network (TCN):** the 168-hour sequence is kept intact, and the features of all plants are combined into one 3D tensor (cases, features, 168 hours). The features are standardised, and the network predicts the whole 168-hour schedule of all 14 generators at once.

4.3 Baseline models

- **v1 (first baseline):** a multi-output Random Forest (Breiman, 2001) on the raw inputs, flattened per case (hourly price and inflows over the week plus initial volumes and water values, 3243 columns) to predict the 2352 targets directly. It reached a micro AUC of 0.782 on an early, smaller split of the data, so the score is not directly comparable with the later ones, but it set the starting point.
- **v2–v3:** the same Random Forest on 58 principal components of the raw data (PCA, v2: 0.883), then combined with the first physical features such as fill levels and head (v3: 0.896).
- **v4 (baseline for all feature experiments):** the predictor was switched to XGBoost, first with one model per generator (v4-A: 0.934), then with upstream predictions passed to downstream plants (topological stacking, v4-B: 0.941), and finally with one model per plant with stacked units (v4-C: 0.942). v4-C is variant **C** in Table 5 and the reference that every later feature set had to beat.

4.4 Version history, including discarded options

Table 4 lists every version, the change it introduced, its score and whether the change was kept. As a rule, a change was kept if it improved the local test score by more than the run-to-run noise, which is about 0.0015 micro AUC, without lowering the OOF score. Three kept changes are at or below this threshold and were kept on additional evidence. v10 (+0.0015 over v9) removed the loss of the surrogate on Songa. v14 (P, +0.0015 over N) raised the OOF score (0.9780 to 0.9788) and the public score (0.98811 to 0.98914). v17 (L, +0.0002 over P) raised the OOF score (0.9788 to 0.9811) and mainly improved the weakest unit, Haukeli (0.898 to 0.923), and for Tokke it follows the weekly running hours much better than the surrogate (Section 3.4).

Three observations summarise the development:

- **Features mattered more than the learner or its hyperparameters.** The largest gains came from features that encode the decision problem: the surrogate (+0.015), the constraint correction (+0.014) and scale-free prices with the Haukeli topology fix (+0.009). Hyperparameter tuning never helped: tuning XGBoost per plant (v7/v7b) lowered the score, and a later screening of the TCN (dropout 0.1/0.3, 192 channels, 15/20 epochs, learning rate $2 \cdot 10^{-3}$, 5 instead of 3 seeds) and of the meta-model regularization did not beat the base configuration by more than 0.002. The models are therefore robust to their hyperparameters, and their performance is limited by the information in the features.
- **Several reasonable ideas did not work** and were discarded: averaging the predictions of overlapping cases and per-unit calibration (both lowered the AUC), ordinal targets, a KNN of similar historical weeks, extra water features and a level correction. The best score of v8

Table 4: Development history. Local test = micro AUC on the 80/20 split; Kaggle = public leaderboard score of the versions that were submitted.

Version	Change	Local test	Kaggle public	Outcome
v1	Random Forest, multi-output, raw inputs	0.782*	–	baseline
v2	PCA (58 components) + Random Forest	0.8830	–	kept as step
v3	v2 + physical features (fill level, head)	0.8957	–	kept as step
v4-A	XGBoost, one model per generator	0.9344	–	kept
v4-B	+ topological stacking of upstream predictions	0.9412	–	kept
v4-C	one model per plant, units stacked	0.9423	0.97778	**new baseline**
v5	+ constraint features for all plants, ordinal targets	0.9349	–	discarded
v6	+ constraint features only for Lio/Byrte, cumulative sums	0.9427	–	discarded (noise)
v7 / v7b	XGBoost hyperparameter tuning per plant (random search)	0.9261 / 0.9393	–	discarded
v8	averaging overlapping cases, per-unit calibration; future volumes (not known at the start of the week)	0.9445‡	–	discarded
v9	+ optimizer surrogate (SURR); KNN of similar weeks	0.9570 / 0.9544	–	SURR kept, KNN not
v10	surrogate only for the plants where it helps (all but Songa)	0.9585§	0.98155	kept
v11	surrogate corrected by min flow / min volume + constraint features (K)	0.9721§	0.98549	kept
v12	scale-free prices and water values, spill pressure, Haukeli topology fix (N); 2021+ weighted ×2	0.9808§	0.98811	kept
v13	+ extra water features, yearly weights, level correction (W)	0.9807	0.98797	discarded
v14	+ per-unit physics and start/stop costs, Viterbi margins (P)	0.9823	0.98914	kept
v15	XGBoost (P) + TCN, blend of logits (20% TCN)	0.9846	0.99054	kept
v17	+ LP of the whole cascade (L)	0.9825	–	kept
v18	XGBoost (L) + TCN, logistic meta-model per unit	0.9865	**0.99112**	**submitted**
v19	AutoGluon (LightGBM, CatBoost, ExtraTrees) (L) + TCN, blend 20% TCN	0.9847	–	kept
v20	attempt to strengthen the TCN (wider, longer training); reverted to v19; no reweighting	0.9847	**0.99101**†	**submitted**
tuning	TCN (dropout, channels, epochs, learning rate, seeds) and meta-model regularization	±0.002	–	discarded

* v1 was evaluated on an earlier, smaller split and is not directly comparable. † 0.99104 with 2021+ cases weighted ×2. ‡ C + future volumes; averaging and calibration lowered the score.
§ Variant chosen per plant (v10: C or S; v11: C, S or K; v12: S, K or N); Table 5 reports each variant applied to all plants.

(0.9445) came only from the future volumes, i.e. the reservoir volumes observed later in the same week. These volumes are a consequence of the schedule being predicted and are not known when it is made, so they break the rule that all features must be known at the start of the week; the feature was discarded for this reason, and its gain (+0.002) was close to the noise anyway.

- **v20 is identical to v19.** v20 started as an attempt to improve the TCN with a wider network and longer training. It did not improve the local score, and the systematic screening above confirmed that training longer slightly hurts (−0.0015 at 15–20 epochs). v20 was therefore reverted to the v19 configuration and submitted under the name v20; the only difference is that its training cases are not reweighted (Section 4.1).

4.5 Contribution of the feature groups

Table 5 isolates the feature variants with the same XGBoost setup. Not every step produced a measurable gain: differences between runs of the same model are about 0.0015 micro AUC, so the last two steps (per-unit physics and the LP) are within the run-to-run variation on the test set. The clearest gains come from the surrogate, the constraint-corrected surrogate and the scale-free price features. Only the step from C to S isolates one change; the later steps each bundle several changes (Table 2), such as the scale-free prices, the spill pressure features and the Haukeli topology fix in N, so their gains cannot be attributed to a single feature group. The gains are also not uniform across units: the surrogate improved Tokke (+0.038 on the test set), Lio (+0.034) and Hogga (+0.020), but made Songa worse (−0.026).

Table 5: Micro AUC of the feature variants (XGBoost, one model per plant, same variant for all plants), trained on the first 80% of the cases and tested after a 7-day gap. The scores can differ slightly from Table 4, which reports per-plant choices of variant (v10–v12) and, for v4-C, an earlier run (0.9423 against 0.9438 here, within the run-to-run noise).

Variant	Content	OOF	Test
C	Base: price, inflow, volumes, water values, stacking	0.9612	0.9438
S	C + surrogate dispatch	0.9716	0.9570
K	S + surrogate corrected by min flow / min volume, constraint features	0.9754	0.9715
N	K + scale-free prices and water values, spill pressure, Haukeli topology fix	0.9780	0.9806
P	N + per-unit physics, start/stop costs, feeder reservoirs	0.9788	0.9823
L	P + LP of the cascade	0.9811	0.9825

The engineered features did what the exploratory analysis suggested. The raw price was a weak predictor of commitment, but the combination of price, water value and reservoir state, summarised by the surrogate, Viterbi and LP features, supports a micro AUC of about 0.983 for the tree models alone on the local test set. The benefit differs by unit: units in the lower cascade and the large Tokke and Vinje stations score around 0.98 to 0.99 with the tree models alone, while Haukeli (about 0.92) and Lio (about 0.94) remain the hardest. Both depend on reservoirs whose water values come from neighbouring reservoirs in our mapping, so the water-value features are probably the least precise for them. This is a plausible explanation rather than something we tested directly, although the topology fix and the LP both mainly improved Haukeli (the LP raised its test AUC from 0.898 to 0.923).

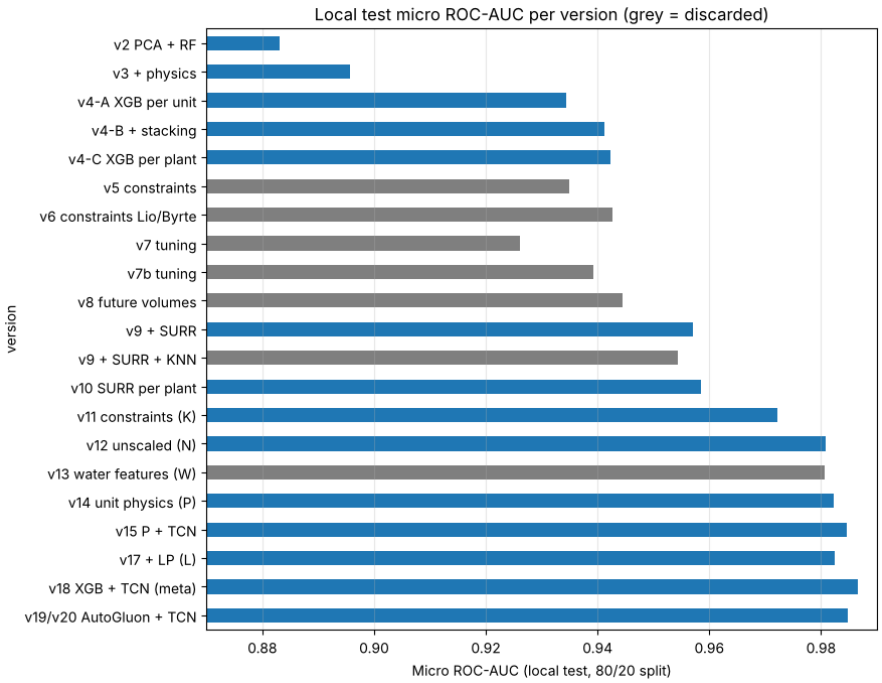


Figure 7: Local test micro AUC of all versions (grey = change discarded).

5 Final Models

The two submitted models use the same feature set (variant L, Section 3) and the same TCN, and differ in the tree-based learner and in how the two predictions are blended (Table 6).

Table 6: The two submitted models.

	Version 18	Version 20
Tree-based learner	XGBoost, one model per plant	AutoGluon: LightGBM, CatBoost, ExtraTrees, one predictor per plant
Neural network	TCN (all plants at once)	same TCN
Blending	logistic meta-model per unit on the logits	fixed weighted average of the logits (20% TCN)
Local test: trees / TCN / blend	0.9825 / 0.9723 / 0.9865	0.9830 / 0.9723 / 0.9847
Weight of 2021+ cases	2.0	1.0 (all cases equal)
Kaggle public score	0.99112	0.99101

5.1 Tree-based models

How the hyperparameters were chosen. The hyperparameters of the tree-based models were fixed by hand, by comparing a few configurations on the local test set (80/20 split, Section 4.1); as described in Section 4.4, a systematic per-plant tuning (v7/v7b) did not improve them. The TCN configuration was instead checked with an internal validation set, the last 20% of the training cases: each variant was trained with one seed and kept only if it improved the validation AUC by more than 0.002, so the local test set was not used to choose the TCN settings. The blending weights (Section 5.3) were chosen on the local test set.

XGBoost (Chen and Guestrin, 2016) (Version 18).

- **Number of trees (`n_estimators`):** 200 (a moderate number of trees to avoid overfitting).

- **Learning rate:** 0.05 (small steps for stable convergence).
- **Maximum depth:** 6 (good results over all folds).
- **Row and column sampling (`subsample`, `colsample_bytree`):** 0.8 (reduces overfitting, as many features are highly correlated).
- **Tree method:** `hist` (histogram-based algorithm for faster training).

AutoGluon (Erickson et al., 2020) (Version 20). To make the second submission more diverse than Version 18, the AutoGluon pipeline trains LightGBM, CatBoost and ExtraTrees instead of XGBoost, with fixed hyperparameters and a weighted ensemble on top:

- **LightGBM (Ke et al., 2017):** 300 boosting rounds, learning rate 0.05, at most 63 leaves, feature fraction 0.5, bagging fraction 0.8.
- **CatBoost (Prokhorenkova et al., 2018):** 300 iterations, learning rate 0.05, depth 6.
- **ExtraTrees (Geurts et al., 2006):** 100 estimators, maximum depth 16, at least 20 samples per leaf, maximum feature fraction 0.3.

Out-of-fold predictions and topological stacking. One model is trained per power plant, in topological order: upstream plants are trained before the plants downstream of them, so that their predicted commitment and expected release can be used as features of the downstream plants (Section 3.5). For the training cases, these features are the out-of-fold predictions of the upstream models (5 contiguous folds of cases). If in-sample predictions were used instead, they would be almost perfect on the training data, and the downstream model would learn to trust them far more than it can at test time, where the upstream predictions are noisier. Out-of-fold predictions have the same noise level in training and at test time.

5.2 Neural network

Trained on the same features (without the stacking features), the neural network alone does not outperform the tree-based models (0.972 against 0.983 on the local test set), but blending the two gives a clear improvement, because the network sees all plants and all hours of the week at once and makes different errors. As neural network, a **Temporal Convolutional Network (TCN)** (Bai et al., 2018) was chosen because it fits the fixed 168-hour window. The convolutions are non-causal, since the whole week of inputs is known in advance.

The architecture and training settings, identical in both versions:

- **Epochs:** 10 (training longer slightly lowered the score).
- **Batch size:** 32.
- **Channels:** 128.
- **Learning rate:** 0.001.
- **Dropout rate:** 0.2.
- **Dilation factors:** 1, 2, 4, 8, 24, 48 and 72, so that the network can reach the same hour on the previous and following three days.
- **Optimizer:** AdamW (Loshchilov and Hutter, 2019) (weight decay 0.01) with a OneCycle learning-rate schedule (Smith and Topin, 2017).

- **Data augmentation:** prices and water values multiplied by 0.5, 1, 2 and 4 during training, so that the network depends on relative rather than absolute price levels.
- **Ensemble seeds:** 3 (three random seeds, averaged).

5.3 Blending

The probabilities of the tree-based model and of the TCN are transformed into logits and combined: in Version 20 with a fixed weighted average (20% TCN), in Version 18 with a logistic regression meta-model per unit (stacked generalization (Wolpert, 1992)), whose weights therefore differ by generator (Section 6.1). The blending weights were chosen on the local test set to maximise the ROC-AUC, and the blended logits are transformed back into probabilities with a sigmoid. Because the weights were chosen on the local test set, the local blend scores are slightly optimistic.

6 Model Interpretation

To understand how the models make unit commitment decisions and how far their scores can be trusted on unseen data, we applied several interpretation methods on the local 80/20 split: per-unit performance, TreeSHAP (global and local) (Lundberg and Lee, 2017; Lundberg et al., 2020), permutation feature importance (Breiman, 2001; Molnar, 2022), partial dependence and individual conditional expectation plots (PDP/ICE) (Friedman, 2001; Goldstein et al., 2015), global surrogate decision trees (Molnar, 2022), Friedman's H-statistic (Friedman and Popescu, 2008) and input gradients for the TCN (Simonyan et al., 2014).

6.1 Performance by generator

Performance varies across generators (Figure 8). Most units reach a ROC-AUC of 0.98 or more, and Byrte, Songa and Vesle Kjela are close to 0.995. Haukeli and Lio are clearly the hardest units, at about 0.95 with the Version 18 meta-model and 0.94–0.95 with the Version 20 blend; the TCN alone is especially weak for Lio (about 0.85).

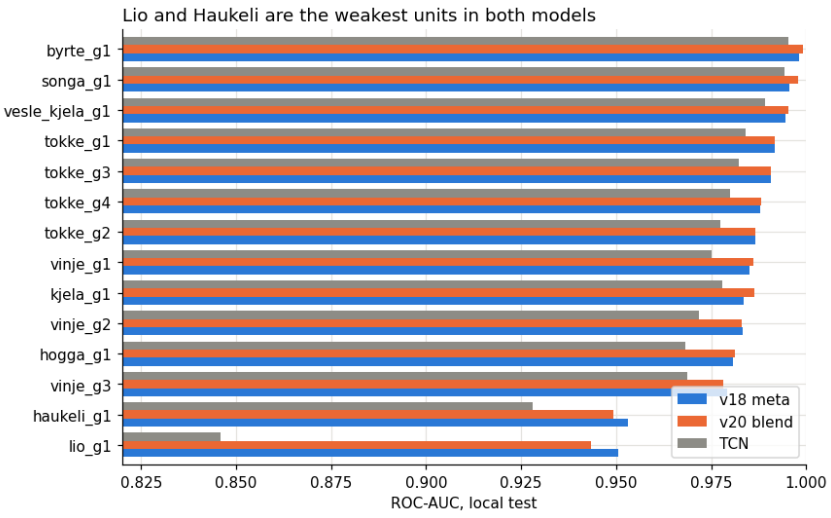


Figure 8: ROC-AUC per generator on the local test set: Version 18 meta-model, Version 20 blend and TCN alone.

The final predictions rely on the TCN to a different degree for each generator (Figure 9). The blending weight is fixed at 0.2 for Version 20, while the meta-model weights of Version 18 differ strongly by unit: the TCN gets the largest share for Haukeli and Songa and the smallest for Tokke, whose tree predictions are already very accurate.

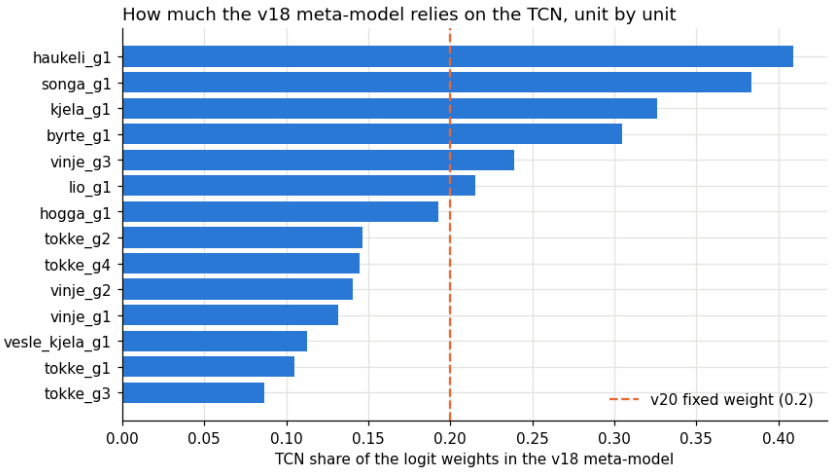


Figure 9: Share of the logit weights given to the TCN by the Version 18 meta-model, per unit; the dashed line is the fixed weight of Version 20 (0.2).

6.2 Global feature importance (TreeSHAP and permutation)

We focus first on the largest plant in the cascade, Tokke, with the XGBoost model of Version 18. The TreeSHAP analysis (Figure 10) shows that individual predictions on the log-odds scale are most strongly driven by three features: the budget-constrained Viterbi margin (`vit_budget`), which accounts for start-up and shutdown costs as well as the available weekly water budget; the estimated upstream water release (`up_release`); and the reduced cost from the cascade LP (`lp_reduced_cost`).

When aggregated by feature group, the five most important groups are, in descending order:

1. Unit physics (including the Viterbi margins).
2. Weekly water budgets and cost-efficient hours (surrogate).
3. Water released by upstream plants (stacking).
4. Cascade LP.
5. Evolution of electricity prices and water values.

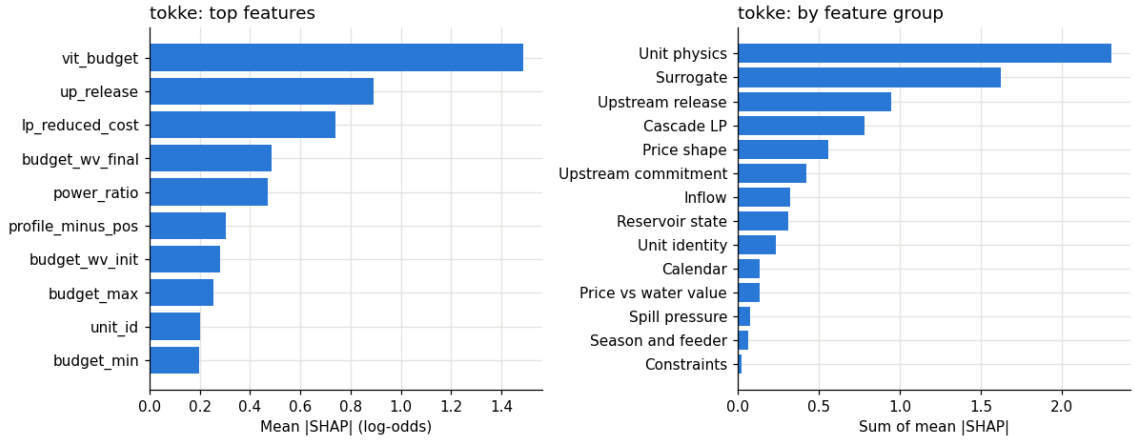


Figure 10: TreeSHAP for Tokke (Version 18, XGBoost): mean |SHAP| of the top features (left) and summed by feature group (right).

Permutation importance (drop in AUC when a feature group is shuffled) confirms this picture and shows that the XGBoost (v18) and AutoGluon (v20) branches learn almost the same feature hierarchy:

- **Engineered features matter more than raw prices.** Each plant is governed by its role in the cascade: upstream plants such as Byrte, Vesle Kjela and Songa are driven mainly by unit physics; Vinje by the spread between price and water value (AUC drop 0.220 / 0.161 for v18 / v20); Haukeli by the cascade LP (0.137 / 0.135); and Lio by the environmental flow constraints (0.164 / 0.169).
- **Correlated features compensate for each other.** For multi-unit plants like Tokke, TreeSHAP shows large individual contributions, but shuffling any single feature group causes only a small drop in AUC (at most 0.020 in v18 and 0.048 in v20), because the engineered features are highly correlated and another group takes over when one is shuffled.

6.3 Local explanation: the worst test week for Tokke

The week with the highest prediction error for Tokke in the local test set shows a failure mode caused by the heavy reliance on the physical features. While earlier models switched units on and off too often, in this week the final model predicts too few shutdowns. Since Generator 1 is usually the first unit to run, we focus on it. The SHAP values (Figure 11) show that the unit-physics features, in particular the Viterbi margins, strongly favour continuous operation, so the model keeps the unit on instead of capturing the shutdowns in the second half of the week.

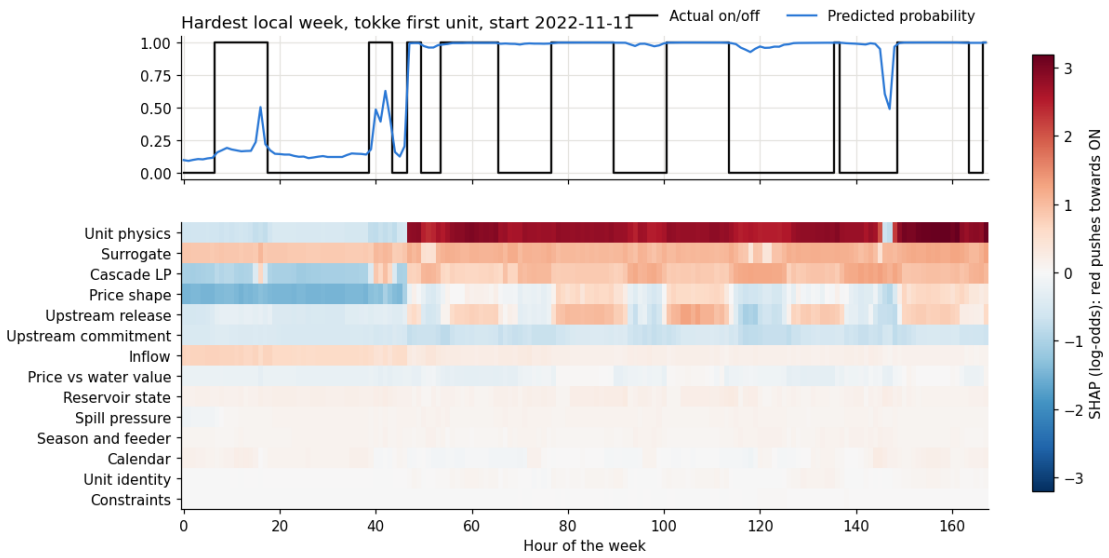


Figure 11: Hardest local test week for Tokke G1 (start 2022-11-11). Top: actual on/off and predicted probability. Bottom: SHAP contribution of each feature group per hour (red pushes towards ON).

At hour 90 of this week the generator is off, but both models predict that it keeps running (Figure 12). The local SHAP analysis shows that both models are driven mainly by unit physics, followed by the surrogate, the reservoir state and the cascade LP. These groups favour operation and outweigh the negative contributions of the price relative to the water value and of the calendar features. The models therefore do not give enough weight to these conflicting signals, which may come from overfitting to the physical features or from an inaccurate representation of the unit's behaviour in unusual periods.

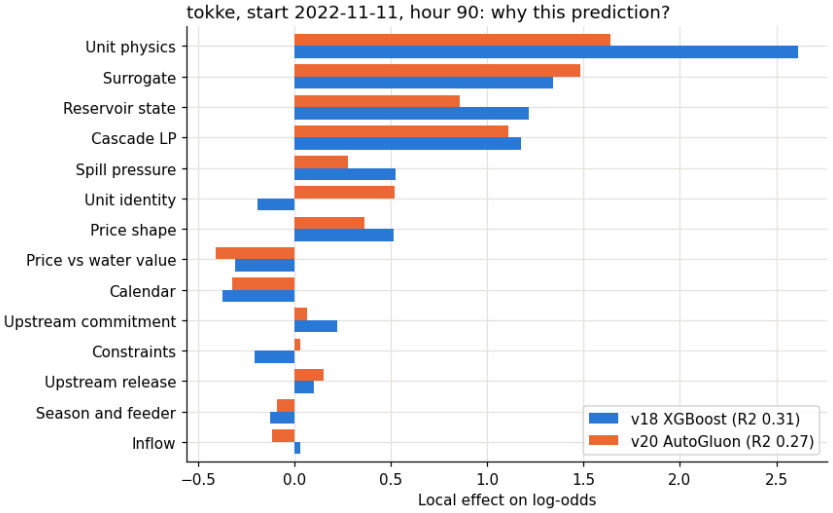


Figure 12: Local effect of each feature group on the log-odds for Tokke G1 at hour 90 of the week starting 2022-11-11 (v18 XGBoost and v20 AutoGluon).

In this week the model also predicts the wrong number of units running in the plant (Figure 13), which is consistent with the earlier finding that most of the error is in "how many" units run rather than "which" units.

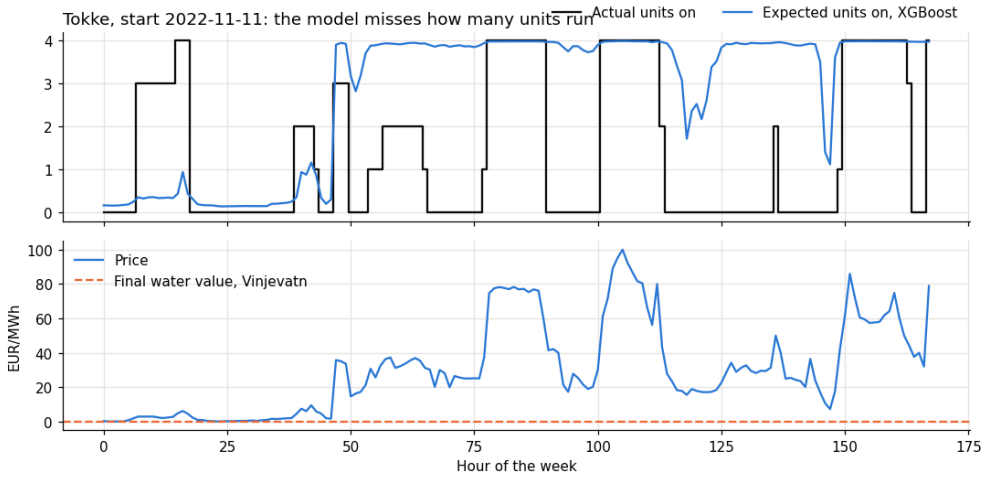


Figure 13: Same week: actual and expected number of running Tokke units (top), day-ahead price and final water value of Vinjevatn (bottom).

6.4 Learned decision rules: PDP/ICE, surrogate trees and interactions

Figure 14 shows the effect of three important features on the predicted log-odds of operation for Tokke, Lio and Haukeli: the average effect (partial dependence, blue) and the responses of individual observations (ICE, grey).

- **Threshold behaviour.** For Tokke, `vit_budget` shows a sharp step around zero, after which the log-odds reach a plateau: both models strongly associate a positive budget margin with operation. The step is considerably stronger for XGBoost than for AutoGluon, which may explain the tendency to predict continuous operation despite actual shutdowns (Section 6.3). For Haukeli, the log-odds rise steeply once `lp_reduced_cost` exceeds zero; for Lio, the price relative to the water value (`log_price_wv_source`) shows a gradual increase followed by a plateau.

- **Both learners agree.** XGBoost and AutoGluon learn the same shapes but differ in sensitivity, with XGBoost reacting more strongly.

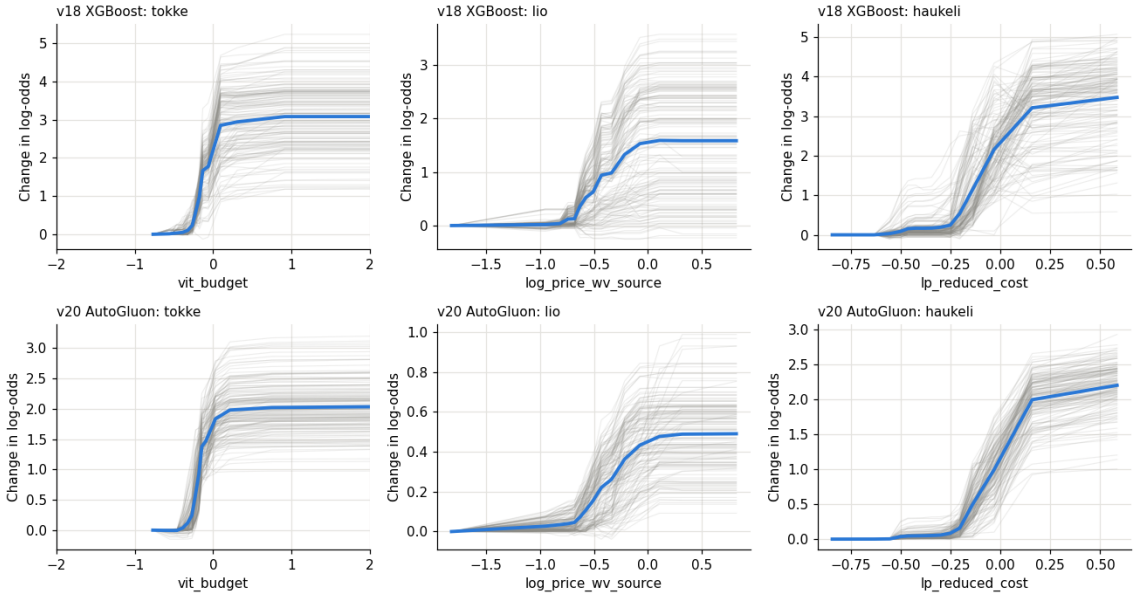


Figure 14: Partial dependence (blue) and ICE curves (grey) for Tokke (`vit_budget`), Lio (`log_price_wv_source`) and Haukeli (`lp_reduced_cost`); top row v18 XGBoost, bottom row v20 AutoGluon.

Global surrogate trees and the H-statistic confirm that the models approximate the logic of the underlying optimization:

- **Optimization thresholds.** The first split of AutoGluon's surrogate tree for Haukeli is `lp_reduced_cost <= 0.00`, the same threshold seen in the PDP.
- **Constraint rules.** For Lio, the surrogate trees of both v18 and v20 split on `need_release` (the flow deficit in Vest-Vassdraget) near 2.3 m^3/s: the models learn to switch the generator on to meet the mandatory minimum river flow even in hours when the price is not profitable.
- **Simple global structure.** A shallow surrogate tree with 8 rules explains most of the variance of the ensembles' log-odds, with R^2 (v18 / v20) of 0.928 / 0.931 for Tokke, 0.887 / 0.917 for Lio and 0.828 / 0.855 for Haukeli. Friedman's H-statistic shows very little pairwise interaction on the log-odds scale ($H^2 \leq 0.018$), most likely because the non-linear physical interactions are already captured by the surrogate, Viterbi and LP features.

6.5 Interpreting the TCN

Permutation importance by feature group gives a ranking for the TCN that is largely consistent with the tree models:

1. Unit physics (including the Viterbi margins): AUC drop 0.052.
2. Weekly water budgets and cost-efficient hours (surrogate): 0.034.
3. Cascade LP: 0.032.
4. Price versus water value: about 0.024.
5. Price shape: 0.020.

The remaining groups (constraints, reservoir state, inflow, spill pressure, calendar) each cause a drop below 0.005.

Because the TCN evaluates all plants at once, it does not use the upstream stacking features of the tree models, and the price versus water value group plays a larger role than in the tree-based models. The network also uses information across plants (Figure 15): shuffling the inputs of Tokke lowers the AUC of Vinje by 0.212 (against 0.066 when shuffling Vinje's own inputs) and of Haukeli by 0.106 (against 0.115 for its own inputs). The network therefore uses the downstream plant, Tokke, to infer the dispatch of the upstream plants that feed it.

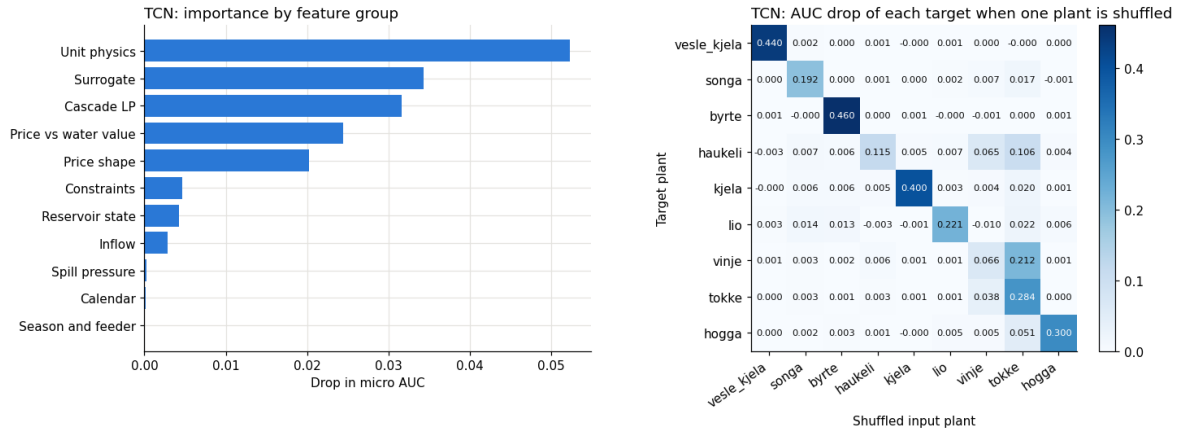


Figure 15: TCN permutation importance by feature group (left) and AUC drop of each target plant when the inputs of one plant are shuffled (right).

The input gradients (Figure 16) show how far in time an hourly prediction looks. The network is by far most sensitive to the inputs at the prediction hour itself. It looks both backward and forward in a roughly symmetric way, as expected from non-causal convolutions over a week whose inputs are all known in advance. The smaller, sharp peaks at offsets of ±1, 2, 4, 8, 24, 48 and 72 hours coincide exactly with the dilation factors of the convolution blocks, so they are mainly an artifact of the architecture: they show where the network is able to look, not a market pattern at those lags. The dilations at 24, 48 and 72 hours were chosen so that the network can reach the same hour on neighbouring days (and the daily-updated volumes and water values), and the plot shows that this reach is used, but with weights an order of magnitude smaller than those of the current hour.

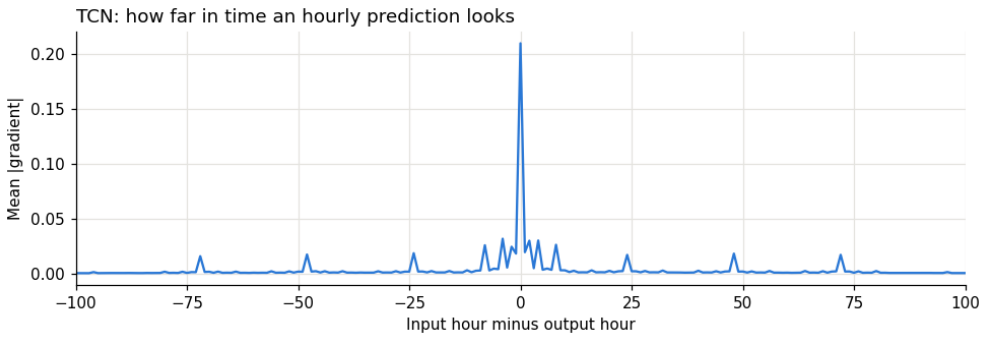


Figure 16: Mean absolute input gradient of the TCN as a function of the offset between input hour and output hour.

6.6 Robustness and limitations

- **Leaderboard stability.** Version 18 (0.99112) and Version 20 (0.99101) perform almost identically on the public leaderboard (part of the 2023–2024 cases). The difference of 0.00011 is

negligible compared with the natural variation between periods on the local test set (splitting it into halves gives a median AUC gap of 0.0020 and a maximum of 0.0054). Both models also dip in the same quarters (2021Q3 and 2022Q2).

- **Shift between the local test set and the Kaggle test set.** The local test set (mid-2021 to 2022) covers the European energy crisis: 94.1% of its weekly mean prices fall outside the 2015–2021 training range. In contrast, none of the 2023–2024 Kaggle cases falls outside the historical price range, as the 2024 quarterly median prices (38–59 EUR/MWh) are closer to pre-crisis levels than those of 2023 (62–105 EUR/MWh). However, 13.2% of the Kaggle weeks (10.4% in 2023, 16.1% in 2024) show extreme relative intraday price swings (standard deviation / mean > 0.616), compared with 0.5% in training and 3.3% in the local test set. On the 19 local test cases with such volatility, the AUC of the Version 20 blend drops to 0.9411.
- **Known failure mode.** Tokke's AUC drops from about 0.99 to 0.928 (v18) and 0.950 (v20) whenever the water value of the upstream Vinjevatn reservoir falls below 5 EUR/MWh, a condition that occurs in roughly 11% of the Kaggle cases.
- **Methodological limitations.** These results describe what the fitted models rely on, not physical causality: a high importance mainly reflects how well a feature approximates the decisions of the SHOP optimizer. Because many features are correlated, they can share or swap credit between runs (for instance, surrogate-tree splits swapping between `net_head` and `langeidvatn_volume`). Finally, the tree hyperparameters, the blending weights and the meta-model regularization were chosen on the local test set, so the local scores of the tree models and of the ensembles are slightly optimistic (the TCN settings were chosen on a separate validation set).

7 Conclusion

We predicted the hourly unit commitment of the 14 generators of the Tokke-Vinje system from prices, inflows, reservoir states, water values and constraints. The EDA showed that the price alone is a weak predictor and that the decisions depend on the comparison between price and water value, on the available water and on the coupling of the cascade. Turning this knowledge into features, in particular three small optimization models (dispatch surrogate, Viterbi start/stop margins and a cascade LP), raised the local micro AUC from 0.942 (XGBoost baseline v4-C) to 0.983 for the tree models alone, while hyperparameter tuning never gave a gain beyond noise. Blending the tree models with a TCN gave the two final submissions, Version 18 (XGBoost + TCN, public score 0.99112) and Version 20 (AutoGluon + TCN, without reweighting, 0.99101). Model interpretation confirms that the models rely mostly on the optimization-based features, and identifies Haukeli and Lio, periods of high intraday price volatility and very low water values at Vinjevatn as the main remaining weaknesses.

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Implementation: scikit-learn (Pedregosa et al., 2011) (Random Forest, PCA, metrics), PyTorch (Paszke et al., 2019) (TCN), SciPy/HiGHS (Huangfu and Hall, 2018; Virtanen et al., 2020) (LP), XGBoost (Chen and Guestrin, 2016) and AutoGluon (Erickson et al., 2020).